

Chapter 3

Properties of Curve Shortening Flow

First, we prove some critical lemmas for the CSF.

Lemma 1. *Let s parametrize γ by arclength. Then as operators we have*

$$\frac{\partial}{\partial s} = \frac{1}{v} \frac{\partial}{\partial u},$$

where $v = \|\frac{\partial \gamma}{\partial u}\|$.

Proof. By definition of arclength, we have:

$$\frac{\partial s}{\partial u} = \frac{\partial}{\partial u} \int_0^u \|\frac{\partial \gamma}{\partial u'}\| du' = \|\frac{\partial \gamma}{\partial u}\| = v.$$

Thus,

$$\frac{\partial}{\partial s} = \frac{\partial u}{\partial s} \frac{\partial}{\partial u} = \left(\frac{\partial s}{\partial u} \right)^{-1} \frac{\partial}{\partial u} = \frac{1}{v} \frac{\partial}{\partial u}.$$

□

Lemma 2. *If $\gamma(u, t)$ evolves under curve shortening flow, then*

$$\frac{\partial v}{\partial t} = -\kappa^2 v.$$

Proof. Here we use the Serret-Frenet equations:

$$\begin{aligned}\frac{\partial T}{\partial u} &= v\kappa n, \\ \frac{\partial n}{\partial u} &= -v\kappa T,\end{aligned}$$

where T is the unit tangent vector to γ . Here, we note that $\frac{\partial}{\partial u}$ and $\frac{\partial}{\partial t}$ commute. Thus, we can differentiate v^2 :

$$\frac{\partial}{\partial t}(v^2) = 2\frac{\partial v}{\partial t}v.$$

and

$$\begin{aligned}\frac{\partial}{\partial t}(v^2) &= \frac{\partial}{\partial t}\left(\left|\frac{\partial \gamma}{\partial u}\right|^2\right) = 2\frac{\partial}{\partial u}\left(\frac{\partial \gamma}{\partial t} \cdot \frac{\partial \gamma}{\partial u}\right) \\ &= 2\frac{\partial}{\partial u}\left(\frac{\partial}{\partial t}\frac{\partial \gamma}{\partial u}\right) = 2\frac{\partial}{\partial u}\left(\frac{\partial^2 \gamma}{\partial u \partial t}\right) = 2vt\frac{\partial}{\partial u}(\kappa n) = 2vT\left(\frac{\partial \kappa}{\partial u}n - v\kappa^2 T\right)\end{aligned}$$

Thus since the normal vector is orthogonal to the tangent vector,

$$2\frac{\partial v}{\partial t}v = -2v^2\kappa^2 \implies \frac{\partial v}{\partial t} = -\kappa^2 v$$

□

Lemma 3. *Under the CSF, as operators, we have that*

$$\frac{\partial^2}{\partial t \partial s} = \frac{\partial^2}{\partial s \partial t} + \kappa^2 \frac{\partial}{\partial s}.$$

Proof. We straightaway compute:

$$\begin{aligned}\frac{\partial^2}{\partial t \partial s} &= \frac{\partial}{\partial t}\left(\frac{\partial}{\partial s}\right) = \frac{\partial}{\partial t}\left(\frac{1}{v}\frac{\partial}{\partial u}\right) = \kappa^2 \frac{1}{v}\frac{\partial}{\partial u} + \frac{1}{v}\frac{\partial}{\partial t}\frac{\partial}{\partial u} = \kappa^2 \frac{1}{v}\frac{\partial}{\partial u} + \frac{1}{v}\frac{\partial}{\partial u}\frac{\partial}{\partial t} \\ &= \kappa^2 \frac{\partial}{\partial s} + \frac{\partial}{\partial s}\frac{\partial}{\partial t} = \frac{\partial^2}{\partial s \partial t} + \kappa^2 \frac{\partial}{\partial s}\end{aligned}$$

□

Lemma 4. *If $\gamma(u, t)$ evolves under curve shortening flow, then*

$$\frac{\partial t}{\partial t} = \frac{\partial \kappa}{\partial s} n$$

and

$$\frac{\partial n}{\partial t} = -\frac{\partial \kappa}{\partial s} T$$

Proof. By the definition of the unit tangent T :

$$\frac{\partial T}{\partial t} = \frac{\partial^2 \gamma}{\partial t \partial s} = \frac{\partial^2 \gamma}{\partial s \partial t} + \kappa^2 \frac{\partial \gamma}{\partial s} = \frac{\partial(\kappa n)}{\partial s} + \kappa^2 T = \frac{\partial \kappa}{\partial s} n + \frac{\partial n}{\partial s} \kappa + \kappa^2 T = \frac{\partial \kappa}{\partial s} n + (-\kappa T) \kappa + \kappa^2 T = \frac{\partial \kappa}{\partial s} n$$

For the second equation, we take the inner product of the tangent with the normal and differentiate with respect to t :

$$T \cdot n = 0 \implies 0 = \frac{\partial t}{\partial t} \cdot n + t \cdot \frac{\partial n}{\partial t} \implies \frac{\partial n}{\partial t} = -\frac{\partial \kappa}{\partial s} t$$

□

Lemma 5. *The curvature changes with time following the equation*

$$\frac{\partial \kappa}{\partial t} = \frac{\partial^2 \kappa}{\partial s^2} + \kappa^3$$

Proof. Here, we let n be the unit normal vector given by an anticlockwise rotation of the unit tangent vector by an angle of $\frac{\pi}{2}$, and take κ to be the curvature induced by this choice of normal vector. Let $\theta(s, t)$ to be the angle between the unit tangent vector $\gamma'(s, t)$ and the x -axis. Then, in coordinates, we have

$$T = (\cos \theta, \sin \theta)$$

Differentiating,

$$\frac{\partial T}{\partial t} = \frac{\partial \theta}{\partial t} (-\sin \theta, \cos \theta) = \frac{\partial \theta}{\partial t} n$$

But as we have shown

$$\frac{\partial t}{\partial t} = \frac{\partial \kappa}{\partial s} n$$

Thus

$$\frac{\partial \theta}{\partial t} = \frac{\partial \kappa}{\partial s}$$

and similarly

$$\frac{\partial T}{\partial s} = \frac{\partial \theta}{\partial s} n$$

And we already know

$$\frac{\partial t}{\partial s} = \kappa n.$$

Thus,

$$\frac{\partial \kappa}{\partial t} = \frac{\partial^2 \theta}{\partial t \partial s} = \frac{\partial^2 \theta}{\partial s \partial t} + \kappa^2 \frac{\partial \theta}{\partial s} = \frac{\partial}{\partial s} \frac{\partial \theta}{\partial t} + \kappa^2 \frac{\partial \theta}{\partial s} = \frac{\partial^2 \kappa}{\partial s^2} + \kappa^3$$

□

Finally, two more key lemmas.

Lemma 6. *Let γ be a closed embedded curve which evolves by curve shortening flow. Let $L(t)$ be the length of γ_t and $A(t)$ the area it encloses. Then:*

- (i) $\frac{dL}{dt} = - \int_{\gamma_t} \kappa^2 ds$,
- (ii) $\frac{dA}{dt} = -2\pi$.

Proof. For (i), we simply use the definition of length:

$$\frac{dL}{dt} = \frac{d}{dt} \int_0^{2\pi} v du = \int_0^{2\pi} \frac{\partial v}{\partial t} du$$

Since $\frac{\partial v}{\partial t} = -\kappa^2 v$, we get:

$$\frac{dL}{dt} = - \int_0^{2\pi} \kappa^2 v du = - \int_{\gamma_t} \kappa^2 ds$$

To prove (ii), let $\gamma(u, t) = (\gamma_1(u, t), \gamma_2(u, t))$ and choose n to be the normal vector pointing inwards. We then use Green's theorem to get an expression for $A(t)$:

$$A(t) = \iint dA = \frac{1}{2} \int_0^{2\pi} \left(\gamma_1 \frac{\partial \gamma_2}{\partial u} - \gamma_2 \frac{\partial \gamma_1}{\partial u} \right) du = -\frac{1}{2} \int_0^{2\pi} v \gamma \cdot n du$$

We now differentiate both sides with respect to t :

$$\frac{dA}{dt} = -\frac{1}{2} \int_0^{2\pi} \left(\frac{\partial \gamma}{\partial t} v n + \gamma \frac{\partial v}{\partial t} n + \gamma v \frac{\partial n}{\partial t} \right) du = -\frac{1}{2} \int_0^{2\pi} \left(\kappa v - \gamma \kappa^2 v n - \gamma \frac{\partial \kappa}{\partial u} t \right) du$$

$$= -\frac{1}{2} \int_0^{2\pi} (\kappa v - \varphi \kappa^2 v n + \kappa v + \varphi v \kappa^2 n) du = - \int_0^{2\pi} \kappa v du = - \int_{\gamma_t} \kappa ds$$

Since each γ_t is a closed curve, we get:

$$\frac{dA}{dt} = -2\pi.$$

□

These last two lemmas are of special importance. They directly show that under the CSF, both length and enclosed area are decreasing functions with respect to time. Furthermore, the area enclosed by a curve is necessarily non-negative, the flow cannot exist beyond $T = \frac{A_0}{2\pi}$. But, we also know the flow exists until the curve shrinks to a point, at which point it must have zero area. Thus, the flow cannot stop before $T = \frac{A_0}{2\pi}$, and so the maximal T is $T = \frac{A_0}{2\pi}$.

We end this section by stating a concrete form of a result we had shown earlier, namely, that the CSF keeps embedded curves embedded.

Theorem 3.0.1. *Let $\gamma : S^1 \times [0, T_0) \rightarrow \mathbb{R}^2$ be a family of closed curves satisfying equation (2.1). If the initial curve γ_0 is embedded and if there exists $c \in \mathbb{R}$ such that $\kappa(u, t) \leq c$ for all $(u, t) \in S^1 \times [0, T_0)$, then γ_t is an embedded curve for each $t \in [0, T_0)$.*